

PEQUOT LAKES, MN, USA

WMO: 727500

Lat: 46.600N

Lon: 94.317W

Elev: 390

StdP: 96.73

Time Zone: -6.00 (NAC)

Period: 82-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.8	-27.3	-37.4	0.1	-30.4	-34.0	0.2	-26.9	5.5	-13.8	4.9	-13.0	1.3	300	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	31.6	21.0	29.7	20.2	27.7	19.0	22.6	29.5	21.3	27.6	20.1	25.9	1.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	15.5	27.0	19.0	14.4	25.1	17.9	13.5	23.7	68.3	29.6	63.5	27.8	59.3	25.9	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.8	4.3	3.5	-34.7	34.7	2.9	2.7	-36.8	36.6	-38.5	38.2	-40.1	39.8	-42.2	41.7		
(5)			-35.6	24.1	3.0	1.5	-37.7	25.2	-39.4	26.0	-41.1	26.8	-43.2	27.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.5	-12.7	-9.9	-2.9	4.8	12.6	17.5	20.2	19.2	13.0	5.9	-3.2	-11.1
	DBStd	12.96	7.52	7.87	6.23	5.26	4.73	4.01	3.68	3.69	4.67	4.82	5.37	7.78
	HDD10.0	3079	704	558	402	172	26	1	0	0	22	146	395	654
	HDD18.3	5241	962	792	659	406	187	63	22	36	170	387	644	913
	CDD10.0	1076	0	0	0	16	106	225	316	284	111	18	1	0
	CDD18.3	196	0	0	0	1	8	38	79	61	9	0	0	0
	CDH23.3	1982	0	0	0	9	129	401	812	544	82	6	0	0
	CDH26.7	626	0	0	0	1	24	122	292	167	19	1	0	0
Wind	WSAvg	1.5	1.5	1.4	1.7	1.8	1.7	1.5	1.2	1.2	1.5	1.6	1.6	1.4
Precipitation	PrecAvg	710	18	16	34	54	81	109	104	79	80	79	33	22
	PrecMax	855	49	49	89	131	192	190	193	160	180	152	104	54
	PrecMin	529	2	1	9	21	31	37	48	22	19	10	4	7
	PrecStd	85	13	11	22	25	37	40	40	39	34	40	24	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.4	6.8	14.0	24.5	29.3	32.7	36.2	33.0	29.1	23.7	12.4	6.1
		MCWB	1.4	2.6	7.3	14.1	18.8	20.7	23.2	22.4	19.2	15.2	9.5	2.7
	2%	DB	1.7	4.4	9.5	20.8	26.7	30.1	32.3	30.6	25.5	19.5	8.8	2.6
		MCWB	-0.6	1.1	4.6	12.0	16.1	20.1	21.4	21.5	18.2	12.7	6.0	0.2
	5%	DB	-0.3	2.3	7.1	17.3	24.4	27.5	29.9	28.5	23.1	16.7	6.2	0.5
		MCWB	-1.8	-0.1	3.2	9.5	15.1	18.6	20.3	19.9	17.0	11.4	3.9	-1.0
	10%	DB	-1.7	0.6	4.7	14.2	22.0	25.4	27.8	26.4	20.8	13.9	4.1	-0.7
		MCWB	-3.0	-1.2	1.7	7.4	14.0	17.5	19.4	18.8	15.8	9.0	2.0	-1.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	2.9	8.2	16.0	19.6	22.8	24.3	23.9	21.2	15.8	10.4	3.7
		MCDB	4.2	6.2	12.9	21.6	28.2	29.9	33.7	30.7	26.0	21.0	11.8	6.0
	2%	WB	-0.5	1.3	5.4	12.9	17.8	20.9	22.7	22.4	19.6	14.3	6.4	0.4
		MCDB	1.5	3.9	9.3	19.7	24.5	27.6	30.1	28.9	23.5	17.4	8.8	2.2
	5%	WB	-1.5	0.1	3.4	10.4	16.3	19.5	21.4	21.1	18.0	12.1	4.1	-0.8
		MCDB	-0.2	2.2	6.7	16.7	22.6	26.2	28.3	26.8	22.1	16.8	6.1	0.5
	10%	WB	-2.9	-1.1	1.9	7.9	14.9	18.3	20.1	20.0	16.6	9.5	2.2	-1.8
		MCDB	-1.6	0.7	4.6	13.7	20.6	24.2	26.3	25.0	20.1	13.7	4.1	-0.6
Mean Daily Temperature Range	5% DB	MDBR	10.6	10.4	9.9	12.1	13.1	12.2	12.0	11.7	11.1	10.5	7.5	9.3
		MCDBR	9.6	10.7	12.3	17.6	15.9	15.1	14.5	14.1	13.7	15.0	10.2	9.0
		MCWBR	8.2	8.4	8.4	10.4	8.4	7.8	7.0	7.0	8.2	9.2	7.8	7.7
	5% WB	MCDBR	9.3	10.1	11.5	15.3	13.5	13.2	13.3	12.3	11.8	14.1	9.6	8.8
		MCWBR	8.0	8.0	8.1	9.7	7.8	7.5	7.3	6.7	8.0	9.3	7.8	7.7
Clear-Sky Solar Irradiance	taub	0.256	0.286	0.308	0.357	0.377	0.376	0.379	0.386	0.355	0.311	0.283	0.263	
	taud	2.378	2.295	2.386	2.343	2.344	2.377	2.387	2.383	2.448	2.562	2.519	2.405	
	Ebn at Noon	854	887	921	899	888	889	880	860	859	859	821	803	
	Edh at Noon	70	94	101	116	121	117	115	111	95	73	62	61	
All-Sky Solar Radiation	RadAvg	1.59	2.72	3.83	4.81	5.36	5.86	6.23	5.24	3.92	2.41	1.49	1.11	
	RadStd	0.18	0.28	0.38	0.41	0.39	0.39	0.27	0.44	0.29	0.36	0.15	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page